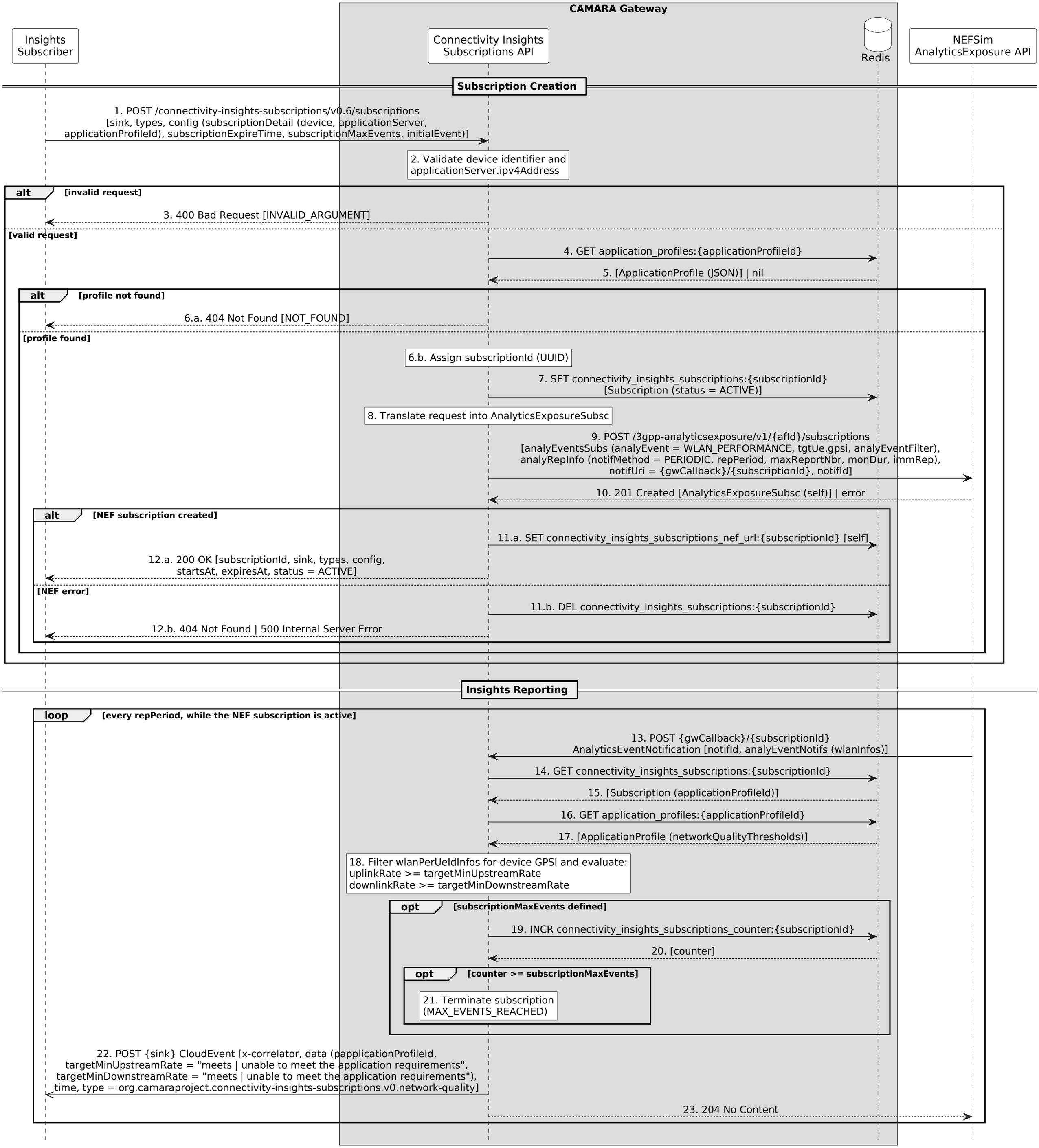
CAMARA Gateway
Insights Subscriber
Connectivity Insights Subscriptions API
Redis
NEFSim AnalyticsExposure API
Subscription Creation
1. POST /connectivity-insights-subscriptions/v0.6/subscriptions [sink, types, config (subscriptionDetail (device, applicationServer, applicationProfileId), subscriptionExpireTime, subscriptionMaxEvents, initialEvent)]
2. Validate device identifier and applicationServer.ipv4Address
alt [invalid request]
3. 400 Bad Request [INVALID_ARGUMENT]
[valid request]
4. GET application_profiles:{applicationProfileId}
5. [ApplicationProfile (JSON)] | nil
alt [profile not found]
6.a. 404 Not Found [NOT_FOUND]
[profile found]
6.b. Assign subscriptionId (UUID)
7. SET connectivity_insights_subscriptions:{subscriptionId} [Subscription (status = ACTIVE)]
8. Translate request into AnalyticsExposureSubsc
9. POST /3gpp-analyticsexposure/v1/{afId}/subscriptions [analyEventsSubs (analyEvent = WLAN_PERFORMANCE, tgtUe.gpsi, analyEventFilter), analyRepInfo (notifMethod = PERIODIC, repPeriod, maxReportNbr, monDur, immRep), notifUri = {gwCallback}/{subscriptionId}, notifId]
10. 201 Created [AnalyticsExposureSubsc (self)] | error
alt [NEF subscription created]
11.a. SET connectivity_insights_subscriptions_nef_url:{subscriptionId} [self]
12.a. 200 OK [subscriptionId, sink, types, config, startsAt, expiresAt, status = ACTIVE]
[NEF error]
11.b. DEL connectivity_insights_subscriptions:{subscriptionId}
12.b. 404 Not Found | 500 Internal Server Error
Insights Reporting
loop [every repPeriod, while the NEF subscription is active]
13. POST {gwCallback}/{subscriptionId} AnalyticsEventNotification [notifId, analyEventNotifs (wlanInfos)]
14. GET connectivity_insights_subscriptions:{subscriptionId}
15. [Subscription (applicationProfileId)]
16. GET application_profiles:{applicationProfileId}
17. [ApplicationProfile (networkQualityThresholds)]
18. Filter wlanPerUeIdInfos for device GPSI and evaluate: uplinkRate >= targetMinUpstreamRate downlinkRate >= targetMinDownstreamRate
opt [subscriptionMaxEvents defined]
19. INCR connectivity_insights_subscriptions_counter:{subscriptionId}
20. [counter]
opt [counter >= subscriptionMaxEvents]
21. Terminate subscription (MAX_EVENTS_REACHED)
22. POST {sink} CloudEvent [x-correlator, data (papplicationProfileId, targetMinUpstreamRate = "meets | unable to meet the application requirements", targetMinDownstreamRate = "meets | unable to meet the application requirements"), time, type = org.camaraproject.connectivity-insights-subscriptions.v0.network-quality]
23. 204 No Content